

1 SAFETY CONSIDERATION AND SANITARY

1.1 Summary

The information in this chapter is designed to instruct persons who own, install, operate, service or inspect this equipment on warning symbols, precautions which must be observed, and explanation of warning labels attached to the machine.



- **Before attempting to perform any operation, maintenance or inspection of this equipment, it is imperative to read and understand the instructions in this manual and to carefully observe all safety precautions and warnings contained herein.**
 - **If there are any unclear points or questions concerning the information contained in this manual, please contact Ishida installation personnel or your nearest Ishida Service representative before proceeding.**
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1.2 Warning Indications: Types and Definitions

The warning indications contained in this manual, as well the indications on the labels attached to the machine, are ranked into three categories according to the level of hazard involved.

It is important that these indications be fully understood and complied with.

Table 1-1 Warning Indications

INDICATION	EXPLANATION
 DANGER	If this hazard is not avoided, death or serious injury will probably result. This indicates a clear and immediate danger, and extreme caution must be exercised to prevent a mishap.
 WARNING	If this hazard is not avoided, there is a possibility of death or injury resulting.
 CAUTION	If this hazard is not avoided, there is a possibility that light or moderate injury may result. It may also indicate that a possibility of damage to equipment exists.
NOTE	Used to emphasize or clarify an important point in the manual.

1.3 General Precautions to be Observed

This section describes the general safety precautions which must be observed when handling this equipment.

DANGER

- All electrical work for the installation site must be performed by licensed electrical contractors.
- All maintenance and inspection work involving electrical components must be performed by qualified maintenance personnel.
Electrical shock or equipment malfunction may result if unqualified personnel are permitted to perform maintenance or inspection of electrical components.

WARNING

- Keep sufficient area around the equipment for operation or maintenance.
- Never touch any electrical switches or buttons with wet or damp hands. Electrical shock may occur when the equipment is not properly grounded or when there is electrical leakage.
- Personnel with long hair using this equipment should tie up their hair securely and all personnel must wear a cap or hat as well as clothes and shoes suitable for the production environment.
Unbound long hair or inappropriate clothing may become caught in moving parts, and injuries may result.
- Before starting operation of the equipment, make sure that all covers are securely shut and fastened.
- To avoid injury, do not touch any moving parts during operation.
- During operation, do not remove the cover of the electrical control unit. There is a possibility of electrical shock.
- Before performing any maintenance or inspection work which is not specifically indicated in this manual, shut off the main power switch.
The person performing the maintenance should lock the main power switch and keep the key in his possession while performing the work.
Injury or electrical shock may result if the equipment is turned on by another person while maintenance work is being performed.
- Turn off the main power switch then wait at least 3 minutes before performing any maintenance or inspection work on the equipment. Residual charge may remain in the machine even after the power has been turned off.
Also, turn off the power of any upstream/downstream units which are connected or adjacent to the equipment. Otherwise there is a danger of electrical shock.
- This equipment is designed with due consideration of sanitation. However, mis-handling may cause an unexpected unsanitary result. Take care to do cleaning.
- If maintenance or inspection work is to be performed with the main power switch ON, clearly indicate this situation by posting a sign in the work area. This is to prevent other personnel from accidentally starting up the equipment.
- When performing maintenance and inspection work on the upper part of the seal checker, use a sturdy ladder or foothold to avoid a hazardous fall.

- Do not use the same outlet as this equipment.
This may cause a fire.



- Make sure to use the power supply specified in the seal checker specification sheet or plate.
If incorrect power supply is used, equipment damage may result.
- This equipment is Class 1 equipment which require protective grounding. To prevent from shock hazard, make sure that a grounding conductor of a power supply cable is firmly grounded.
- Do not apply excessive pressure to the power cord by placing this equipment or any objects on it.
There is possibility of power cord being disconnected.
- Do not power off the seal checker during production except when emergency stop is required,
because the setting or statistics data may be lost.
- Do not run the equipment with any tools or other objects placed on top of the machine.
If objects fall into moving parts, damage to equipment may result.
- Use finger to press the display keys on the operation panel. Using a ball-point pen or other pointed object can damage the operation panel.

1.4 Special Safety Precautions

This section describes special safety precautions for this equipment which, in addition to the previously mentioned precautions, should be carefully observed.

WARNING

- Before cleaning the equipment, shut off the main power switch.
The person performing the cleaning should lock the main power switch and keep the key in his possession while cleaning.
Injury may result if the equipment is turned on by another person while cleaning is being performed.

CAUTION

- Install this equipment in a location fulfilling the following conditions:
 - Ambient temperature range: 0-40°C
 - Ambient Humidity: 30%-85% (no condensation)
 - A hard horizontal surface that does not vibrate
- Do not use the same power source for devices which may emit noise. Doing so may result in malfunction or damage.
- Do not place near wireless or other equipment emitting electronic waves. This may result in malfunction or damage.
- Do not install wiring which permits load-related power fluctuations of more than $\pm 10\%$. Overload may result in malfunction or damage.
- Before attempting to shift this equipment, contact the distributor where the equipment was purchased or an Ishida Service representative.
Unstable equipment may result in inaccurate checking. For this reason, make sure the equipment is securely fastened.
- When cleaning, be sure to rinse only the areas indicated. Otherwise, damage or malfunction may result.
For areas not indicated as washable, clean by wiping with a dry cloth.
- Do not apply insecticide or other foreign substances inside the main body, terminal boxes, motor boxes, or other enclosed seal checker compartments. Doing so may cause machine malfunction or damage.

1.5 Warning Labels

Warning labels which indicate points requiring particular caution are attached to the seal checker at certain locations. Please take sufficient time to familiarize yourself thoroughly with the meanings and positions of these labels.

1.5.1 Warning Label Handling

- First verify that all the warning labels are clearly legible. If the label text or graphic is difficult to read, clean or replace the label.
- Clean labels with water and neutral cleanser.
Do not use organic solvents or gasoline.
- Labels must be replaced if they are damaged, peeled or illegible.
For this, please contact your local Ishida Service representative.

1.5.2 Warning Label Location

The diagram below shows the location of the warning label.

 label shows the location where the electric control unit is accommodated.

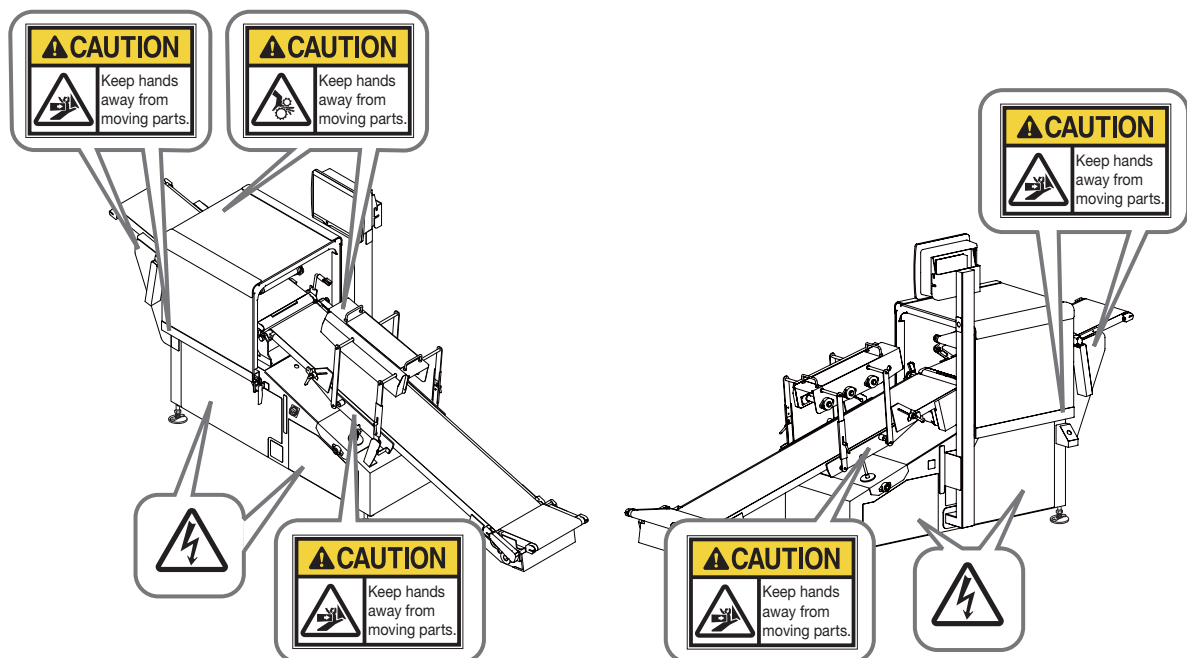


Fig.1-1 Warning Label Location

1.6 Electrical Power Shutdown and Indication

Before performing maintenance or inspection, electrical power should be shutdown to ensure the safety of personnel.

To prevent other personnel from starting operation while this work is being performed, follow the procedure described below.

- The main power switch should be shut off and locked.
- A tag clearly indicating that maintenance work is in progress should be prepared and posted on the power shutdown device.

Lock the main power switch and attach the accident prevention tag as shown below.

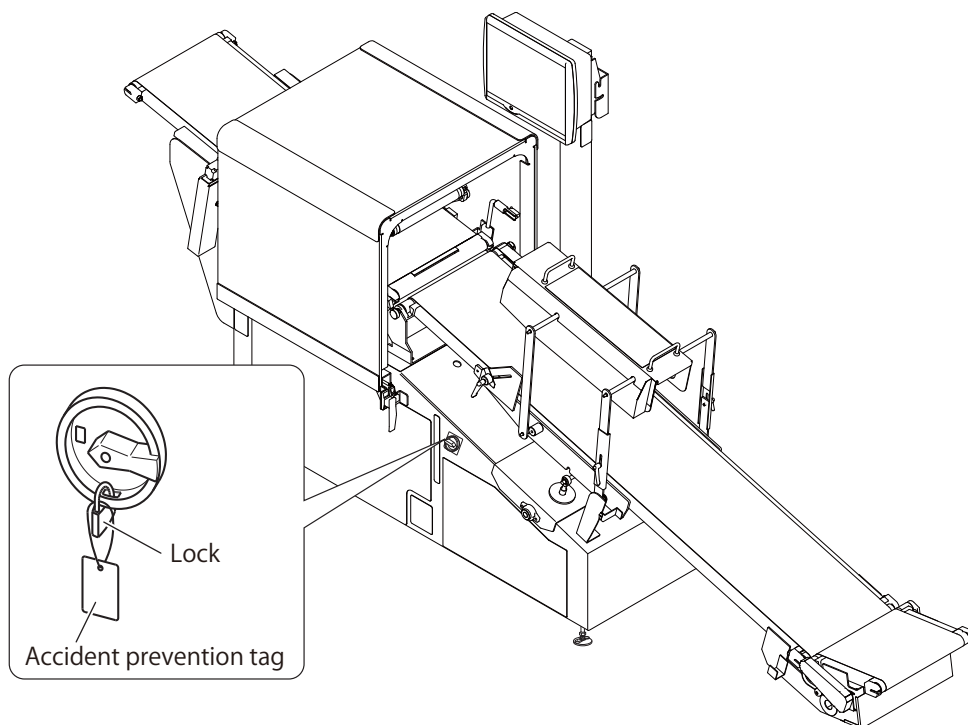


Fig.1-2 Main Power Switch

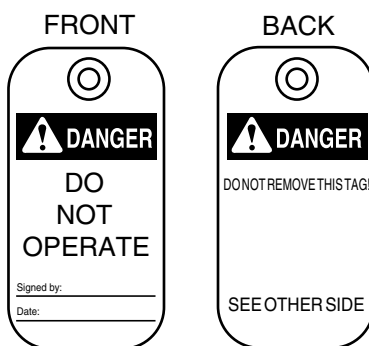


Fig.1-3 Accident Prevention Tag (Example)

